sijantanv@gmail.com

Tanvir Ahmed Sijan

https://sijantanvir.github.io/

/tanvir-sijan

/sijantanvir

Research Interests

My research interests lie at the intersection of NLP and AI safety, focusing on human-AI alignment through human-centered evaluation, alignment methods, and LLM interpretability to develop safer and more trustworthy language models.

Education

Jahangirnagar University — Dhaka, Bangladesh
Master of Science in Computer Science — *May 2025 – Sept. 2026*

- **CGPA**: 3.98/4.00
- **Relevant Coursework**: Machine Learning (A+), Deep Learning (A+), Natural Language Processing (A+), Data Mining (A), Artificial Intelligence & Expert Systems (A+), Computational Intelligence (A+)
- **CS Prerequisites**: Data Structures & Algorithms (A+), Software Engineering & Database Systems (A+), Computer Architecture & Operating Systems (A+), Programming Languages (A+)

Bangladesh University of Engineering and Technology (BUET) — Dhaka, Bangladesh
Bachelor of Science in Civil Engineering, with Honors — *Feb. 2020 – Mar. 2025*

- **CGPA**: 3.77/4.00, **Major**: Transportation Engineering, **Minor**: Environmental Engineering
- **Math and Programming Coursework**: Differential Equations & Statistics (A+), Differential, Integral Calculus and Matrices, Computer Programming Sessional (A+), Engineering Computation Sessional (A+)

Test Scores

GRE: 324 (Quantitative: 168/170, Verbal: 156/170, AWA: 4.0/6.0)
TOEFL iBT: Overall 5.5/6.0 (110/120) (Speaking: 5.5/6.0)

Research Experience

Graduate Researcher — Jun. 2025 – Present
Computational Intelligence, Data and Language Research Lab — *Jahangirnagar University*
Supervisor: Dr. Md. Musfique Anwar

- **Mechanistic Interpretability:** Co-conceptualized a study of arithmetic heuristic neurons across three Llama models (1B/3B/8B), testing (1) whether arithmetic neurons are form-invariant across symbolic, python code, and natural language word-problem formulations and (2) whether cross-format failures reflect distinct circuits or differential activation of a shared circuit; conducted experiments on the 1B-parameter model and co-authored the manuscript. Identified a compact, causally sufficient shared circuit whose activation transfer recovered over 97% of cross-format failures for addition and subtraction, with lower recovery for multiplication and division (BlackboxNLP at EMNLP 2026).
- **Event Extraction:** Led a comprehensive study of event detection from Bangla news, developing a generalized news event ontology, coordinating annotation of 9,979 sentences across 40 subtypes and three noise conditions, and training and evaluating encoder-based and decoder-only models; found decoder-only LLMs markedly more robust to noise, while instruction tuning with annotation guideline improved performance on noisy text but did not consistently improve robustness (W-NUT at EMNLP 2026)
- **Cultural Alignment:** Conceptualized and led *BanglaSocialBench* end-to-end, integrating LLM APIs, building evaluation pipelines, coordinating annotation, and conducting statistical analyses for a 1,719-instance benchmark evaluated across 12 LLMs; uncovered systematic over-politeness, failure to recognize multiple socially acceptable address-pronoun forms, and cross-religious kinship-term conflation (ACL SRW 2026, Best Paper Award).

Research Intern — May 2025 – Oct. 2025
Maks Inc. — *Dhaka, Bangladesh (HQ: Frisco, TX, USA)*

* Conducted literature reviews and collaborated with data scientists and transportation engineers to analyze large-scale BRTA traffic datasets, develop emission models, perform geospatial analyses, and conduct in-person stakeholder interviews across major administrative divisions of Bangladesh; prepared technical reports for projects funded by the **World Bank** and **Asian Development Bank**.

Undergraduate Researcher Nov. 2023 – Feb. 2025

Advisor: Dr. Md. Shamsul Hoque; Research Mentor: Dr. Md. Asif Raihan *BUET*

* Built an end-to-end computer vision and machine learning framework integrating street-view imagery, open geospatial data, and crash records to determine the impact of visual and built-environment characteristics on pedestrian crash severity, applying SHAP-based interpretability techniques.

Peer-Reviewed Publications

** indicates equal contribution*

[1] Sharath Naganna*, Tanvir Ahmed Sijan*, Uddipta Kalita. **"Are Arithmetic Heuristic Neurons Form-Invariant? A Mechanistic Analysis of Symbols, Text, and Code in LLMs."** *Proceedings of the 9th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP, co-located with EMNLP 2026*, October, 2026. [ACL Anthology] [Code]

[2] Tanvir Ahmed Sijan, S. M Golam Rifat, Nayeemul Islam, Md. Musfique Anwar. **"Beyond Clean Text: Evaluating Encoder and Decoder Robustness for Bangla Event Detection in Noisy Text."** *Proceedings of the 11th Workshop on Natural User-generated Text, co-located with EMNLP 2026*, October, 2026. [ACL Anthology] [Code]

[3] Tanvir Ahmed Sijan, S. M Golam Rifat, Pankaj Chowdhury Partha, Md. Tanjeed Islam, Md. Musfique Anwar. **"BanglaSocialBench: A Benchmark for Evaluating Sociopragmatic and Cultural Alignment of LLMs in Bangladeshi Social Interaction."** *ACL Student Research Workshop*, 2026. [**Best Paper Award**] [ACL Anthology] [Code] [Video]

[4] Tanvir Ahmed Sijan, Md Asif Raihan, Sk. Md. Mashrur, Md. Shamsul Hoque. **"Assessing the Visual and Built Environment Risk on Pedestrian-Vehicle Crash Severity in Absence of Detailed Roadway Inventory: A Deep Learning Based Framework Harnessing Public Geospatial Data."** *Accepted at 106th Transportation Research Board (TRB) Annual Meeting*, Washington, D.C., 2027. [PDF] [Code]

Honors & Awards

Best Paper Award Jul. 2026

ACL Student Research Workshop 2026

Undergraduate Student Research Grant Jan. 2024

Research and Innovation Centre for Science and Engineering, BUET

- Awarded competitive student research grant for developing a deep learning based framework, incorporating street-view imagery and open geospatial data with SHAP-based interpretability, to model pedestrian-vehicle crash severity.

Dean's List Scholarship Jul. 2019

Bangladesh University of Engineering and Technology (BUET)

Higher Secondary Certificate (HSC) Scholarship 2019

Board of Intermediate and Secondary Education, Rajshahi

- Awarded Talentpool Scholarship; ranked 10th (∼top 0.01%) among all candidates in the Rajshahi Board.

Service and Leadership

Workshop Reviewer Jul. 2026

11th Workshop on Natural User-generated Text (W-NUT) collocated with EMNLP 2026

Deputy Vice President Dec. 2024 – Mar. 2025

ITE BUET Student Chapter

- Collaborated with executive members to organize workshops and seminars on various topics of transportation research.

Vice President Mar. 2024 – Mar. 2025

ACI BUET Student Chapter

- Organized poster competitions, seminars, and field visits in coordination with chapter members.

Skills

Programming: Python, C, C++, SQL, MATLAB
Libraries: PyTorch, NumPy, Pandas, Transformers, Matplotlib, Scikit-Learn
Other: Linux, Git, Bash, LaTeX
Languages: Bangla (Native), English, Hindi (Basic)